Jihwan Hong

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Research Interests

- Video Understanding and Generation
- World Modeling
- 3D/4D Reconstruction and Generation

Education

Seoul National University 2025 – Present
M.S. in Artificial Intelligence

Korea University 2019 – 2025
B.S. in Computer Science and Engineering

Research Experience

Research Assistant, AIDAS Lab, Seoul National University, Korea Aug. 2025 – Present
Advisor: Prof. Jaeyoung Do

Research Intern, AIDAS Lab, Seoul National University, Korea Jan. 2025 – Aug. 2025
Advisor: Prof. Jaeyoung Do

Research Intern, MLV Lab, Korea University Jun. 2023 – Dec. 2024
Advisor: Prof. Hyunwoo J. Kim

Research Intern, AI Lab, Korea University Oct. 2020 – Jan. 2021
Advisor: Prof. Dongsuk Yook

Publications

When Predicting Nothing Beats SAM 3: Revisiting Evaluation in Video Object Segmentation 2026
*40th Annual Conference on Neural Information Processing Systems (**NeurIPS**), Evaluations and Datasets Track*
Jihwan Hong, Woohyeon Park, Jaeik Kim, Jaeyoung Do†
Project · Paper · arXiv

VIRST: Video-Instructed Reasoning Assistant for SpatioTemporal Segmentation 2026
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*
Jihwan Hong, Jaeyoung Do†
Project · Paper · arXiv · Code

Report of the 5th PVUW Challenge: Towards More Diverse Modalities in Pixel-Level Understanding 2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)
Chang Liu *et al.*
Paper · arXiv

Preprints & Manuscripts

Dynin-Omni: Omnimodal Unified Large Diffusion Language Model 2026
Under review

Jaeik Kim, Woojin Kim, **Jihwan Hong**, Yejoon Lee, Sieun Hyeon, Mintaek Lim, Yunseok Han, Dogeun Kim, Hoeun Lee, Hyunggeun Kim, Jaeyoung Do†

Demo · Code · Model · arXiv

DM3SR: Benchmarking Deep Multi-View 3D Spatial Reasoning 2026
Under review

Jusang Oh, **Jihwan Hong**, Jaeyoung Do†

SoccerNet 2024 Challenges Results 2024
arXiv preprint

Anthony Cioppa *et al.*

arXiv

Awards & Honors

3rd Place, 5th PVUW Challenge MeViS-Audio Track 2026
CVPR 2026 Pixel-level Video Understanding in-the-Wild (PVUW) Workshop

3rd Place of MeViS-Audio Track of the 5th PVUW: VIRST-Audio

Jihwan Hong, Jaeyoung Do†

arXiv

2nd Place, CVPRW 2024 SoccerNet Challenge: Multi-View Foul Recognition 2024
CVPR 2024 CVsports Workshop

VF-VARS: Leveraging Video Foundation Models for Video Assistant Referee Systems

Jihwan Hong*, Youngseo Kim*, Junseok Lee, Jimin Lee, Sehyung Kim, Hyunwoo J. Kim†

Technical Report

1st Place, The 14th KSME Student Creative Design Competition 2023
The Korean Society of Mechanical Engineers; Minister of Science and ICT Award, Government of South Korea

Development of an AI-Based Automatic Seat Belt Adjustment Device

Code

Academic Service

Reviewer 2027
International Conference on Learning Representations (ICLR)

Reviewer 2026
IEEE Transactions on Multimedia

Selected Projects

Anatomy- and Action-Conditioned Colonoscopic World Model Jul. 2026 – Present
SNU Boramae Medical Center & NVIDIA HK

Developing a colonoscopic I2V world model conditioned on an anatomical topology map and actions.

Technical Drawing Registration Automation Apr. 2025 – Jun. 2025
POSCO DX

Developed a vision-language-model-based workflow for autonomous technical drawing registration.